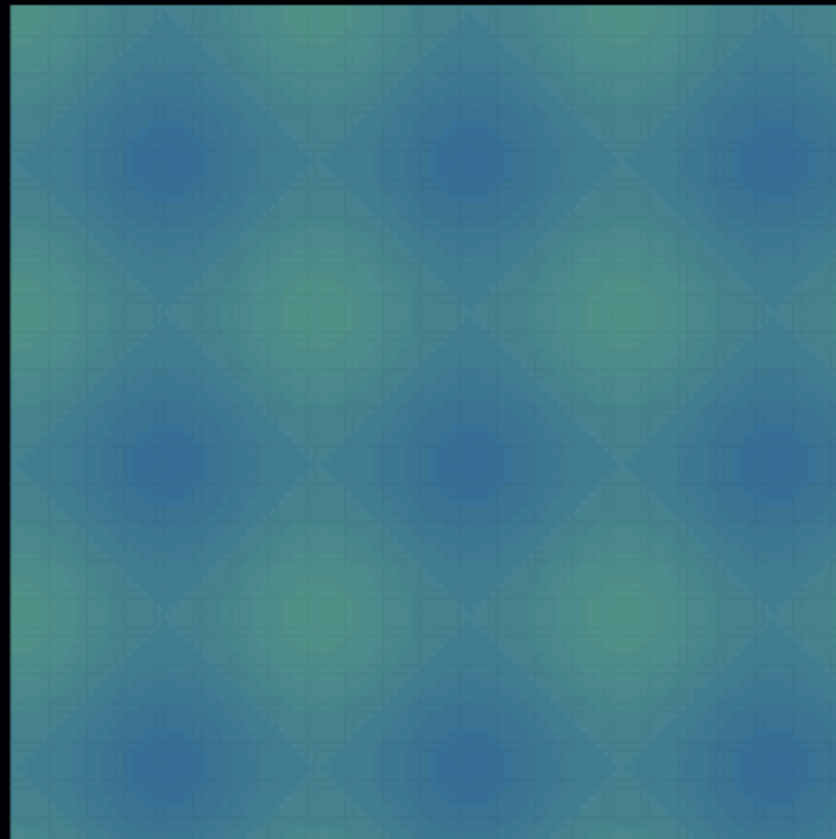


Faraday Waves

Pause

Reset

Parametric (Mathieu) instability of a thin liquid layer driven at ω_d : surface modes oscillate at $\omega_d/2$, snapping into ordered patterns

 $\eta_{L_{max}} = 0.45$ 

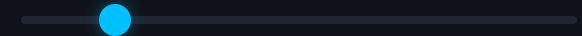
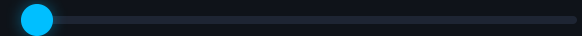
PATTERN

Mode set (number of standing-wave directions)

2 directions (90°) $\rightarrow$ squares

VERTICAL DRIVE

Drive frequency ω_d 5.5 rad/s

Drive amplitude ε 3Damping γ 0.02

Each standing-wave amplitude obeys a damped Mathieu oscillator $A'' + 2\gamma A' + \omega_0^2(1 + \epsilon \cos \omega_d t)A = 0$ with $\omega_0 = \omega_d/2$. Inside the first instability tongue (low γ , ϵ past threshold) the response subharmonically locks at $\omega_d/2$; superposing 1, 2 or 3 standing waves selects stripes, squares or hexagons.

MEASURED VALUES

 ω_d 5.5 rad/s

What sets the pattern (stripes vs squares vs hexagons)?

The selection comes from weakly nonlinear interactions of competing modes — damping, fluid depth, and meniscus boundary conditions favour different symmetries. Our simulator only chooses a few Fourier modes, so the patterns we draw are illustrative cartoons rather than predictions.

How is this related to the swing-pumping demo?

Pumping a swing by squatting at twice the swing frequency is the same parametric resonance: the natural oscillator is excited by a periodic modulation of one of its parameters (effective length here, restoring force there).

Related simulators

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NEW

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Water Hammer (1D)

▶ Launch Simulator

Linearized P,V waves; valve closes; Joukowsky $\Delta P \approx \rho a V$ hint.

NEW

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Rocket Propulsion

▶ Launch Simulator

Variable mass: thrust $\dot{m}u$, Tsiolkovsky Δv , vertical launch with gravity.

NEW

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Kapitza Pendulum

▶ Launch Simulator

Pivot shakes vertically: fast driving can stabilize the inverted equilibrium — parametric pumping in θ' + (g/L)...

NEW

SCHOOL

Coupled Pendulum Chain

▶ Launch Simulator

N pendula with neighbor springs in θ : watch traveling waves and reflections after a center kick.

NEW

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Forced Oscillator

▶ Launch Simulator

Driven damped harmonic oscillator: transients, resonance curve $A(\omega)$.

NEW

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Quarter-Car Suspension

▶ Launch Simulator

¼-vehicle vertical model: sprung vs unsprung masses, K_s , C_s , tire K_t , sinusoidal road — RK4 time histories.